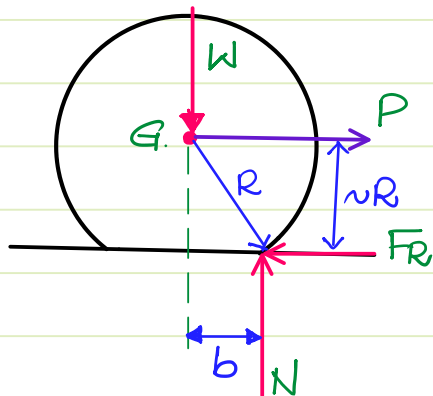


Case 3: Rolling Resistance: The resisting force acting on a body which is trying to roll but is not able to roll is known as Rolling Resistance. In the case of Rolling Resistance the bodies & surfaces are assumed to deform in order to create eccentricity b/w Weight & Normal. There are 3 possible-types of deformation.

1. Rolling body alone is deforming ***
2. Surface is deforming
3. both the rolling body & surface deform.



$$\begin{aligned}\sum F_x = 0 &\Rightarrow F_R = P \\ \sum F_y = 0 &\Rightarrow N = W \\ \sum M = 0 &\Rightarrow \{F_R\}R = W.b \\ &\Rightarrow F_R = \frac{W.b}{R}\end{aligned}$$

F_R : Rolling Resistance
 b : Coeff. of Rolling Resistance.

* Standard Terms In Friction:

1. Angle of friction: $\{\phi\}$ The angle made by Resultant of N & f with the Normal is Angle of friction. $\{\phi\}$

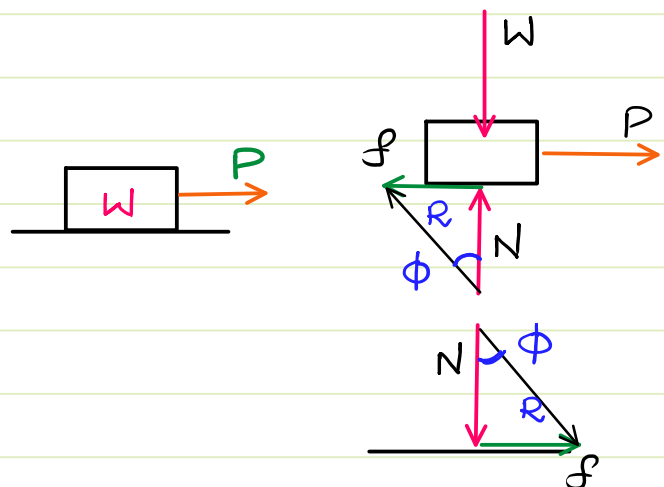
$$\phi = \tan^{-1} \{f/N\}$$

- Angle of static friction:

$$\phi_s = \tan^{-1} \mu_s$$

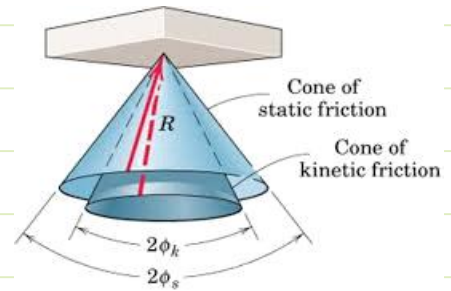
- Angle of kinetic friction:

$$\phi_k = \tan^{-1} \mu_k$$



2. Cone of friction: The cone formed by Normal as height & friction force as radius is known as Cone of friction.

Every possible resultant of Normal & friction is present within the cone of static friction.

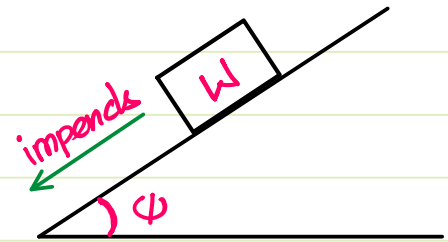


3. Angle of Repose: $\{\psi\}$ The angle of inclination at which a body impends to slide down an incline is known as Angle of Repose.

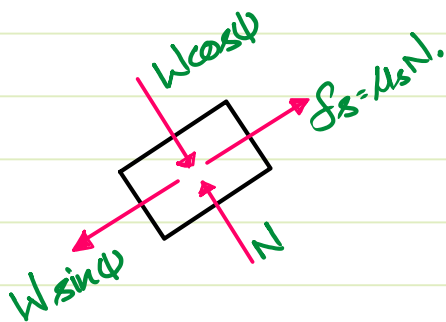
When a body impends motion, that is the last point of eq^m.

Note: In case of particle body

$$\psi = \phi_s$$



Proof:

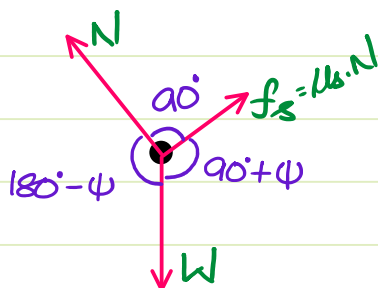


By eq^m: $\sum F_{\perp \text{ to incline}} = 0$
 $\Rightarrow N = W \cos \psi$
 $\Rightarrow f_s = \mu_s \cdot W \cdot \cos \psi$

$\sum F_{\text{along incline}} = 0$
 $f_s = W \sin \psi$
 $\Rightarrow \mu_s \cdot W \cos \psi = W \sin \psi$

$\Rightarrow \mu_s = \tan \psi$ { We also know $\mu_s = \tan \phi_s$ }
 $\therefore \psi = \phi_s$

Alternate Method

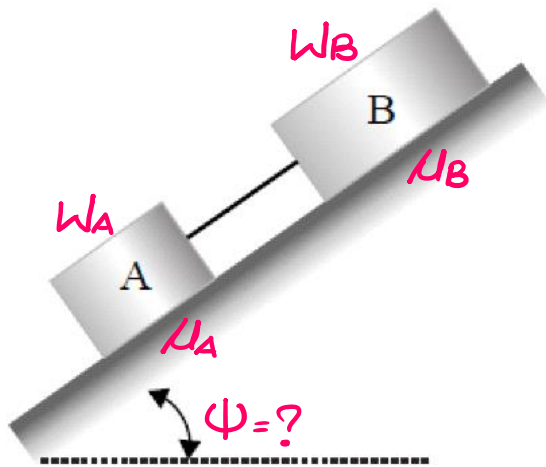


By Lami's:

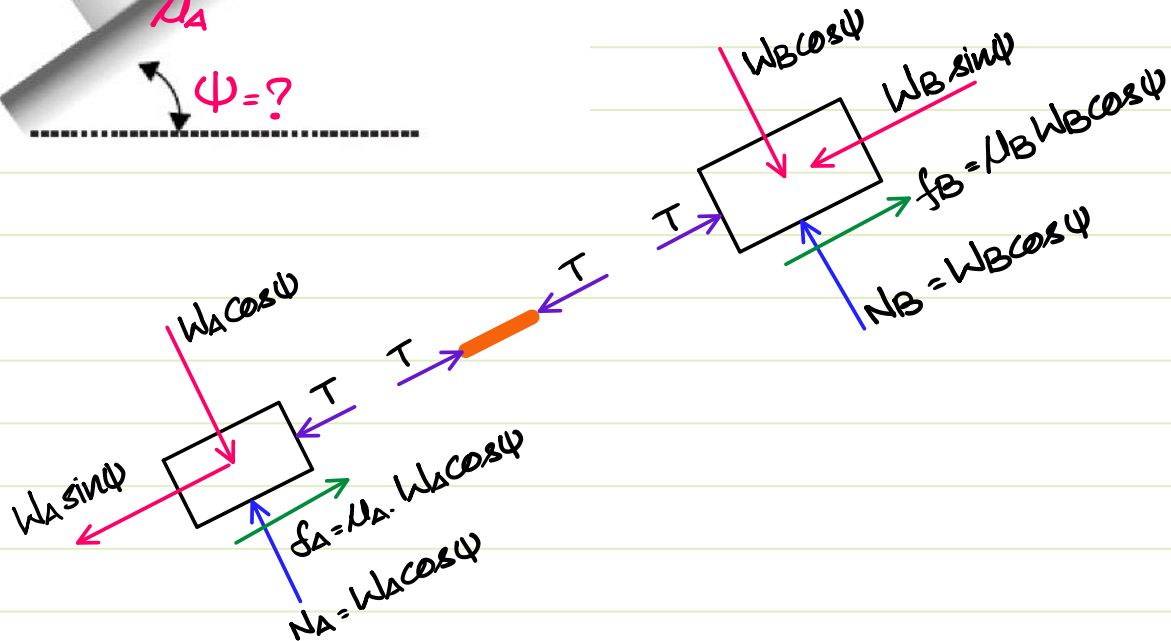
$$\frac{\mu_s N}{\sin [180^\circ - \psi]} = \frac{N}{\sin [90^\circ + \psi]} \Rightarrow \mu_s = \tan \psi$$

$$\Rightarrow \tan \phi_s = \tan \psi$$

$$\psi = \phi_s$$



Given: Two particle bodies $\{W_A, W_B\}$
connected by Massless rigid link
Find: The angle of repose for
the assembly.



$\sum F$ along the incline for the entire system = 0

$$\Rightarrow f_A + f_B = W_A \sin \psi + W_B \sin \psi$$

$$\Rightarrow \cos \psi \{ \mu_A W_A + \mu_B W_B \} = \{ W_A + W_B \} \sin \psi$$

$$\Rightarrow \tan \psi = \left\{ \frac{\mu_A W_A + \mu_B W_B}{W_A + W_B} \right\}$$

$$\Rightarrow \psi = \tan^{-1} \left\{ \frac{\mu_A W_A + \mu_B W_B}{W_A + W_B} \right\}$$

Note: If n blocks were connected, then

$$\psi = \tan^{-1} \left\{ \frac{\sum_{i=1}^n (\mu_i W_i)}{\sum_{i=1}^n W_i} \right\}$$

1 Mark Which among the following statements is true for a body moving on a dry surface under the action of applied forces?

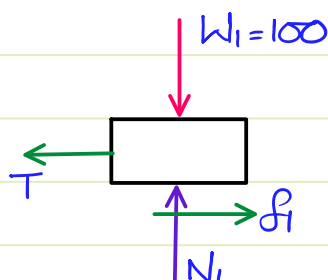
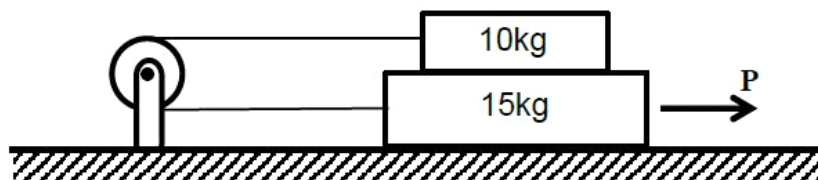
- (A) Kinetic-friction force is zero.
- (B) Kinetic-friction force is equal to the static-friction force.
- (C) Kinetic-friction force is greater than the static-friction force.
- ✓ (D) Kinetic-friction force is lower than the static-friction force.

2 Marks.

Two rigid blocks, of masses **10kg** and **15kg**, are arranged one on top of the other and placed on a horizontal rough surface as shown. The blocks are connected to each other through an inextensible cable passing over a frictionless pulley. The coefficients of static friction between the blocks and also between the bottom block and the surface are all equal to **0.3**. The force **P** (in **Newtons**) needed to set the blocks in motion towards right is _____.

⇒ $\Sigma \vec{F} = 0$ is valid.

(Assume acceleration due to gravity $g = 10\text{m/s}^2$).

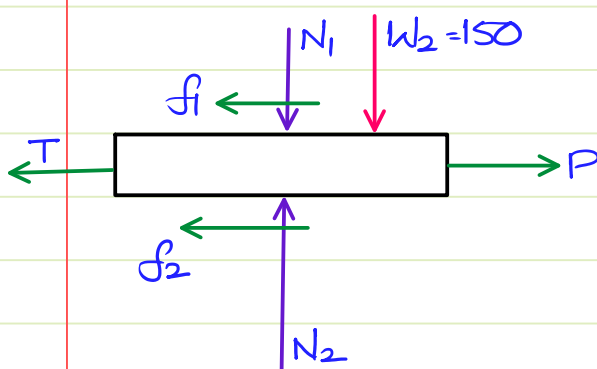


For the 10kg block:

$$\Sigma F_y = 0 \Rightarrow N_1 = 100$$

$$f_1 = \mu N_1 = 30\text{N}.$$

$$\Sigma F_x = 0 \Rightarrow T = f_1 = 30\text{N}.$$



For 15kg block:

$$\Sigma F_y = 0 \Rightarrow N_2 = N_1 + W_2$$

$$= 250\text{N}.$$

$$f_2 = \mu N_2 = 75\text{N}.$$

$$\Sigma F_x = 0 \Rightarrow P = f_1 + T + f_2$$

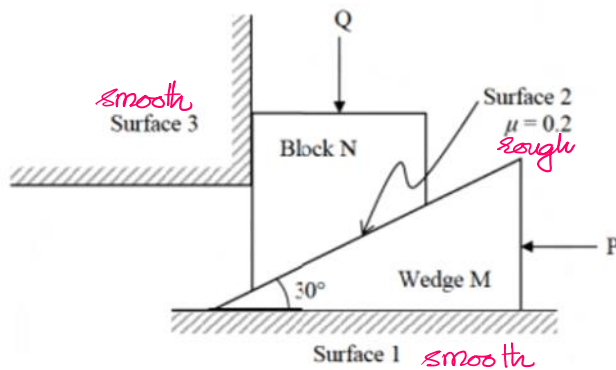
$$\therefore \boxed{P = 135\text{N}.$$

Wedge Friction: A simple Machine used for lifting or moving heavy loads by applying a relatively smaller force in a convenient direction.

- Height of wedge is neglected in calculations.
- Normal & friction can be reduced to resultant force.

2 Marks

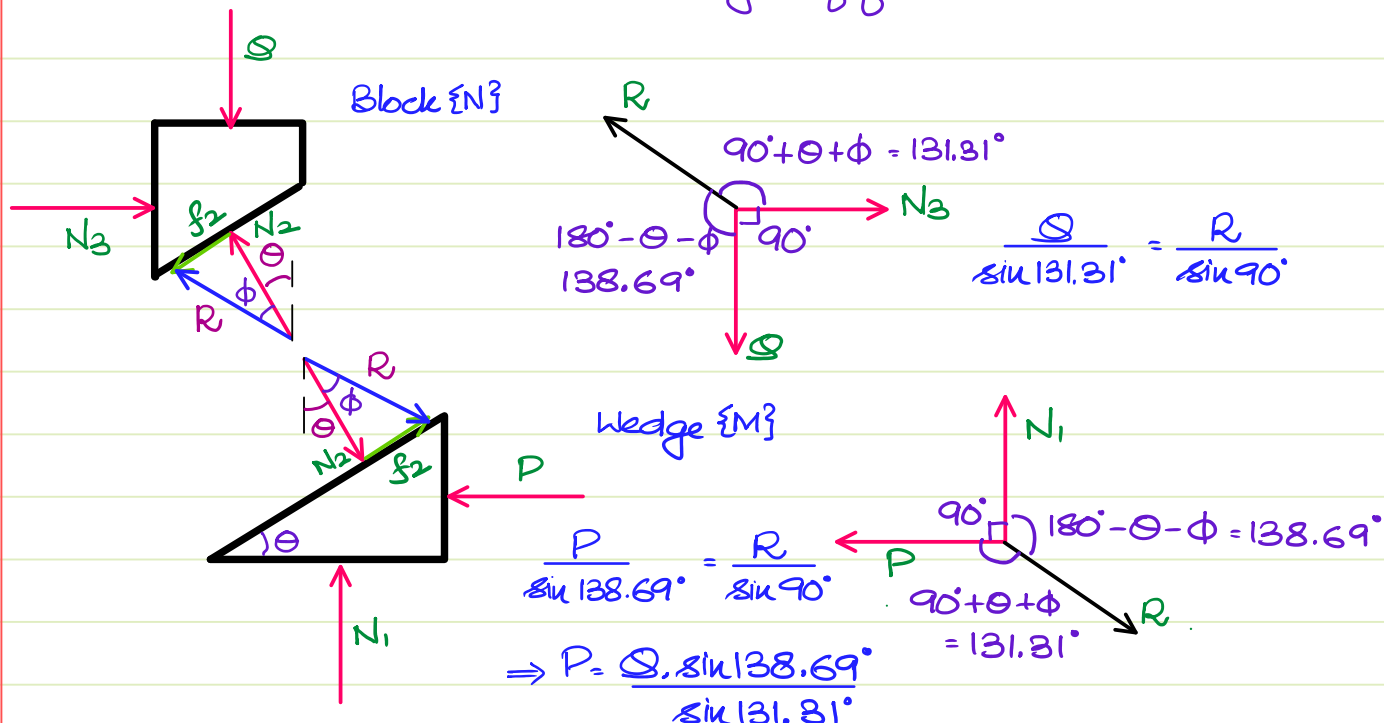
A wedge M and a block N are subjected to forces P and Q as shown in the figure. If force P is sufficiently large, then the block N can be raised. The weights of the wedge and the block are negligible compared to the forces P and Q. The coefficient of friction (μ) along the inclined surface between the wedge and the block is **0.2**. All other surfaces are frictionless. The wedge angle is **30°**. The limiting force P, in terms of Q, required for impending motion of block N to just move it in the upward direction is given as **P = α Q**.



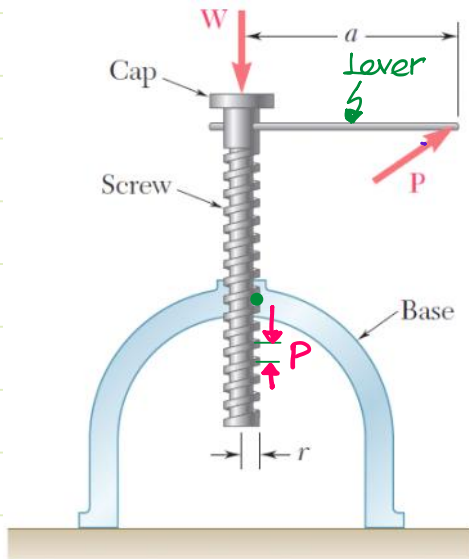
The value of the coefficient ' α ' (round off to one decimal place) is

- (A) 0.6
(B) 0.5
(C) 2.0
☒ (D) 0.9

Angle of friction $\{\phi\} = 11.31^\circ$



Screw Jack: Screw Jack is a machine used for lifting, lowering or holding heavy loads, by applying relatively smaller forces in convenient direction.



W : Load acting on the cap of screw.

P : force applied at the end of lever

r : pitch circle radius.

a : lever arm of the lever.

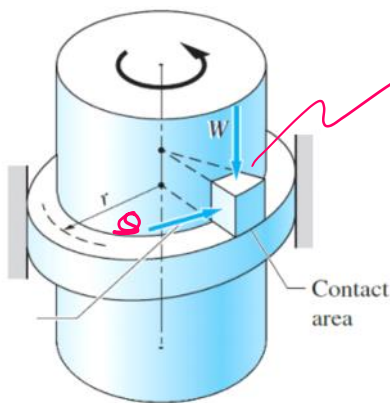
p : pitch of the screw {distance b/w 2 cons. pts on threading of a screw.}

Torque by force " P " abt the axis of screw:
 $T = P \cdot a$

Note: The load acting on the cap of a screw jack is effectively acting on the threading in contact with the base.

S : The effective force-transmitted on the load at the threading due to the torque { $T = P \cdot a$ }

$\Rightarrow$ Torque by P = Torque by S
 $P \cdot a = S \cdot r$

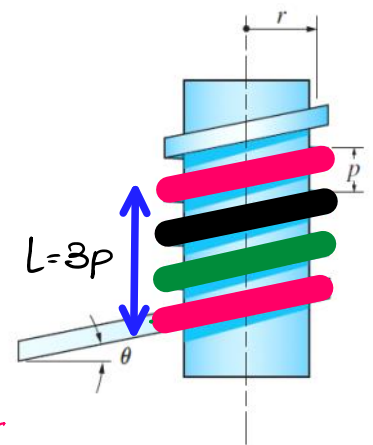


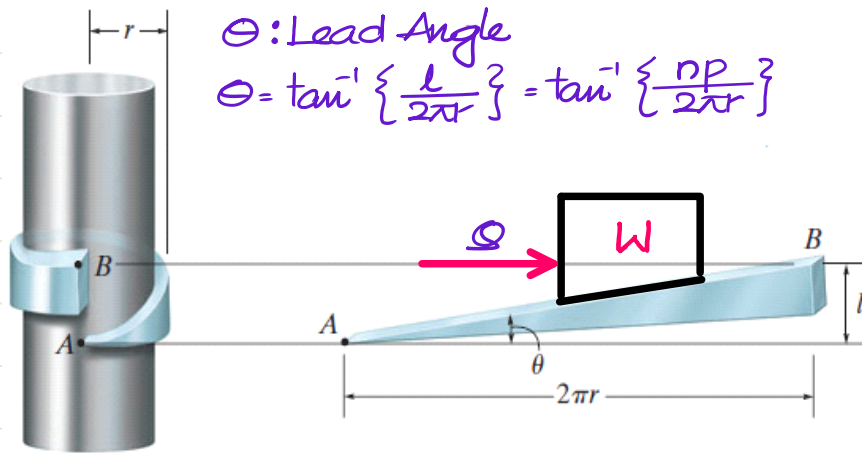
A screw can have multiple starts.

Let the No. of starts {No. of threadings} be equal to n .

The longitudinal advance of the screw upon one complete revolution is given by **Lead** { $l = np$ }

Ex: Triple threaded screw





Θ : Lead Angle
 $\Theta = \tan^{-1} \left\{ \frac{l}{2\pi r} \right\} = \tan^{-1} \left\{ \frac{np}{2\pi r} \right\}$

Unwrapping a screw gives an inclined surface.

⇒ Analysis of a load on screw reduces to load placed on an incline.

There are 2 types of screw:

1. Self Locking Screw:

The load remains at a position to which it is lifted even if the force P is released.

⇒ the friction is sufficient enough to hold the load.

⇒ $\phi_s > \Theta$

1. Lifting the load.

$Q_{\text{lift}} = W \cdot \tan \{\Theta + \phi_s\}$

2. Lowering the load.

$Q_{\text{lower}} = W \cdot \tan \{\phi_s - \Theta\}$

2. Non Self Locking Screw:

The load slides down the incline by itself when the force P is released.

⇒ the friction is not sufficient to hold the load.

⇒ $\phi_s < \Theta$

1. Lifting the load.

$Q_{\text{lift}} = W \cdot \tan \{\Theta + \phi_s\}$

2. Holding the load.

$Q_{\text{hold}} = W \cdot \tan \{\Theta - \phi_s\}$

Efficiency of A screw Jack:

$\eta = \frac{\text{Lifting w/o friction}}{\text{Lifting with friction}}$

$\eta = \frac{\tan \{\Theta\}}{\tan \{\Theta + \phi_s\}}$

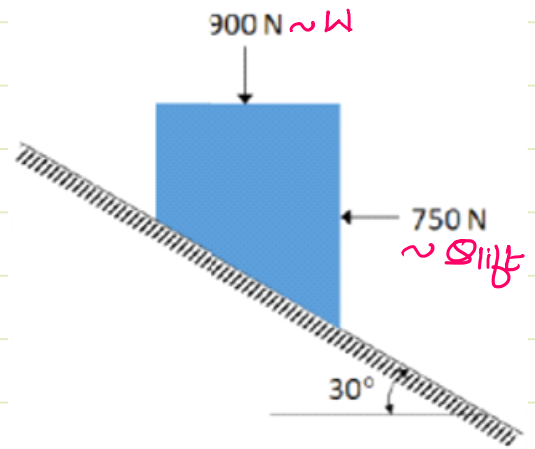
Condition for η_{max} :

$\Theta = 45^\circ - \frac{\phi}{2}$

$\eta_{\text{max}} = \frac{1 - \sin \phi}{1 + \sin \phi}$

2 Mark

A block of negligible mass rests on a surface that is inclined at 30° to the horizontal plane as shown in the figure. When a vertical force of **900 N** and a horizontal force of **750 N** are applied, the block is just about to slide up. The coefficient of static friction between the block and surface is **0.17** (round off to two decimal places).



$$W = 900 \text{ N} ; Q = 750 \text{ N} ; \theta = 30^\circ$$

$$Q = W \cdot \tan \{ \theta + \phi_s \} \Rightarrow 750 = 900 \cdot \tan \{ 30^\circ + \phi_s \}$$

$$\phi_s = \tan^{-1} \left\{ \frac{5}{6} \right\} - 30^\circ$$

$$= 9.8^\circ$$

$$\mu = \tan \phi_s = 0.1728$$

A screw jack has square threads with **75 mm mean diameter** and **15 mm pitch**. The load on the jack revolves with the screw. The coefficient of friction at the screw threads is 0.05. Find the tangential force applied to the jack at 360 mm radius so as to lift a load of 6000 N. State whether the jack is self locking. If it is, find the torque required to lower the load. If not, find the torque which must be applied to keep the load from descending.

$$\phi_s = \tan^{-1} \mu_s = 2.862^\circ$$

$$\theta = \tan^{-1} \left\{ \frac{DP}{2\pi r} \right\} = \tan^{-1} \left\{ \frac{15}{75\pi} \right\} = 3.642^\circ$$

$\therefore \theta > \phi_s \therefore$ Non self locking screw jack

$$Q_{\text{lifting}} = W \cdot \tan \{ \theta + \phi_s \} = 684 \text{ N.}$$

$$P \cdot a = Q \cdot r \Rightarrow P_{\text{lifting}} = Q_{\text{lifting}} \cdot \frac{r}{a} = \frac{684 \times 37.5}{360}$$

$$\therefore P_{\text{lifting}} = 71.25 \text{ N.}$$

$$T_{\text{hold}} = Q_{\text{hold}} \cdot r = W \cdot \tan \{ \theta - \phi_s \} \times 0.0375$$

$$= 3.063 \text{ N.m.}$$

2.4. Virtual Work Method

This is an alternate approach to analyse a system in eq^m if drawing FBD & writing standard eqⁿs of eq^m becomes difficult {Eq^m of multiple connected members like mechanisms & machines}

Principle of Virtual Work: If a body in eq^m under the influence of a set of forces $\{F_1, F_2 \dots F_n\}$ is given an imaginary displacement $\{\delta \vec{r}\}$ then

$$\begin{aligned} \text{Virtual Work: } W &= \vec{F}_1 \cdot \delta \vec{r} + \vec{F}_2 \cdot \delta \vec{r} \dots + \vec{F}_n \cdot \delta \vec{r} \\ &= \{ \vec{F}_1 + \vec{F}_2 \dots + \vec{F}_n \} \cdot \delta \vec{r} \\ &= \sum \vec{F} \cdot \delta \vec{r} \end{aligned}$$

$$\because \sum \vec{F} = 0 \text{ due to eq}^m \quad \therefore \sum \vec{F} \cdot \delta \vec{r} = 0$$

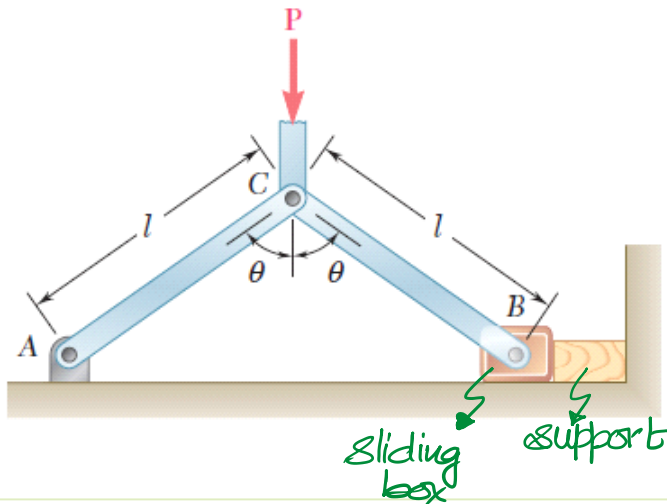
$$\therefore \text{Net Virtual Work} = 0$$

$\because$ the assigned displacement is imaginary, $\therefore$ the work is also imaginary. Hence it is known as Virtual Work.

** Steps Involved In The Analysis of Virtual Work:

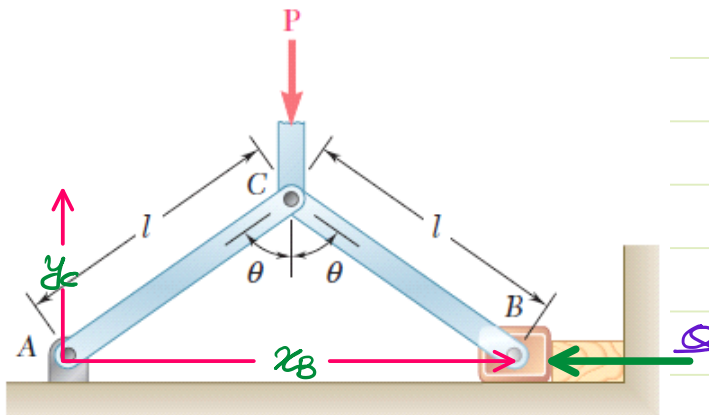
1. Choose a ref point & identify the locations / positions of the forces acting in the system.
2. Assign virtual displacements based on constrain eqⁿ.
3. Write the eqⁿ of virtual work = 0.

Q.



If the crank shaft {AC} & the connecting rod {CB} are of equal length, & the surfaces are smooth, find the reacⁿ from the support onto the sliding box.

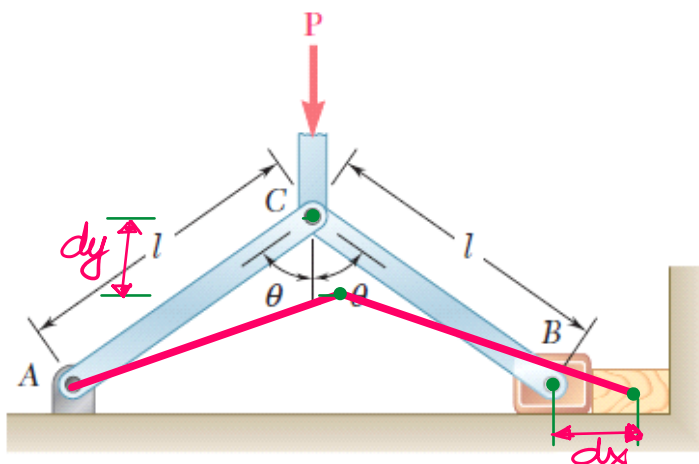
Step 1: Identify the position of forces by taking A as ref. pt.



$$x_B = 2l \sin \theta$$

$$y_C = l \cos \theta$$

Step 2: Assign virtual displacements



$$\frac{dx}{d\theta} = 2l \cos \theta$$

$$\Rightarrow dx = 2l \cos \theta \cdot d\theta$$

$$\frac{dy}{d\theta} = -l \sin \theta$$

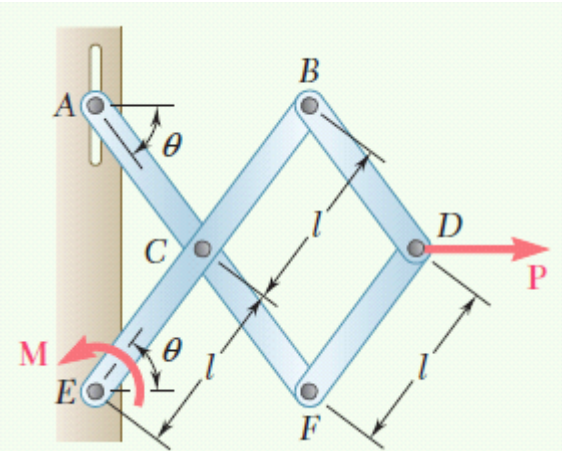
$$\Rightarrow dy = -l \sin \theta \cdot d\theta$$

Step 3: Eqⁿ of Virtual Work

$$P dy + Q dx = 0$$

$$P \{-l \sin \theta, d\theta\} + Q \{2l \cos \theta, d\theta\} = 0$$

$$\Rightarrow Q = \frac{P}{2} \tan \theta$$



find the relationship b/w P & M in order to establish eq^m on the linkage shown. {Neglect friction}

Step 1: Ref pt is E

$$x_D \text{ w.r.t. E} = 3L \cos \theta$$

Step 2: $\frac{dx_D}{d\theta} = -3L \sin \theta$

$$\Rightarrow dx_D = -3L \sin \theta d\theta$$

Step 3: Eqⁿ of Virtual Work:

$$M d\theta + P dx_D = 0$$

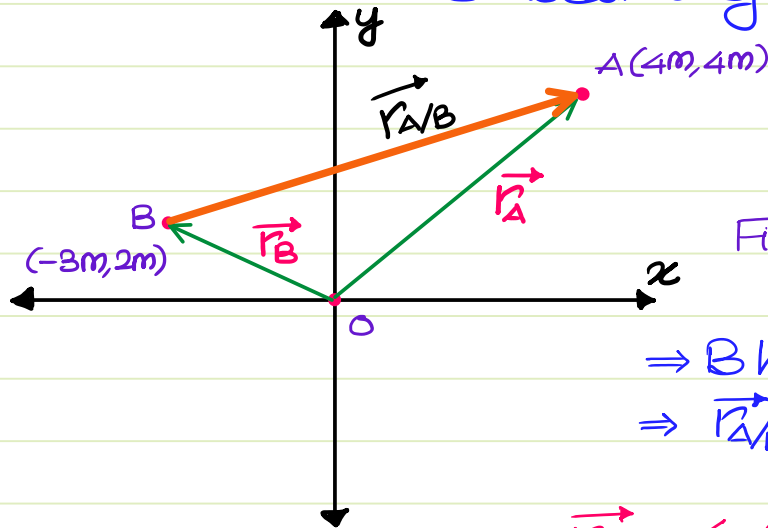
$$M d\theta + P \{-3L \sin \theta, d\theta\} = 0$$

$$\Rightarrow M = 3PL \sin \theta$$

3. Kinematics

3.0. Terminologies

1. Position Vector $\{\vec{r}\}$: A vector which represents the position or location of a body wrt a reference.



$$\vec{r}_A = \{4\hat{i} + 4\hat{j}\} \text{ m}$$

$$\vec{r}_B = \{-3\hat{i} + 2\hat{j}\} \text{ m}$$

Find the position of A w.r.t. B.
 $\vec{r}_{A/B}$

$\Rightarrow$ B has become effective origin

$$\Rightarrow \vec{r}_{A/B} = \vec{r}_A - \vec{r}_B$$

$$= \{4\hat{i} + 4\hat{j}\} - \{-3\hat{i} + 2\hat{j}\}$$

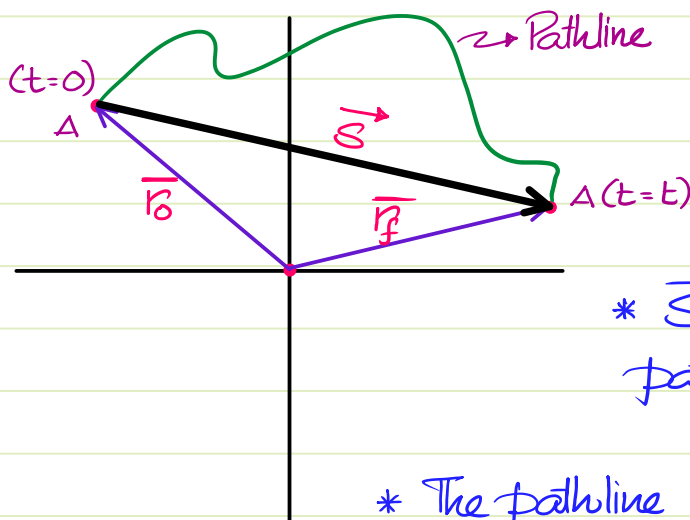
$$\therefore \vec{r}_{A/B} = \{7\hat{i} + 2\hat{j}\} \text{ m.}$$

Note: The position vector is a function of observer.

- * Displacement Vector $\{\vec{s}\}$: Change in position vector of a body.

$$\boxed{\vec{s} = \Delta \vec{r}}$$

SI: m.



* Change in position vector {displacement} is independent of observer.

* $\vec{s}$ is a point function, not a path function.

* The pathline {or} path taken gives a scalar quantity known as distance travelled.

3. Velocity Vector $\{\vec{v}\}$: The rate of change of displacement (or) position of a body wrt time.

Instantaneous Velocity $\{\vec{v}\}$: $\vec{v} = \frac{d\vec{s}}{dt}$ Always tangential to the pathline.

Average Velocity $\{\vec{v}_{av}\}$: $\vec{v}_{av} = \frac{\Delta\vec{s}}{\Delta t}$ Like vector to the displacement.

S.I.: m/s.

If a problem on Average Velocity is asked, always use:
 $\vec{v}_{av} = \frac{\text{Total displacement}}{\text{Total Time}}$

4. Acceleration Vector $\{\vec{a}\}$: Rate of change of Velocity Vector wrt time.

Instantaneous Accⁿ $\{\vec{a}\}$: $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{s}}{dt^2}$

Average Accⁿ $\{\vec{a}_{av}\}$: $\vec{a}_{av} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\text{Total change in Velocity}}{\text{Total Time}}$

S.I.: m/s²

* Rectilinear Motion: Motion of a particle (or) translation w/o rotation of a rigid body along straight line.

* only 1-translational DOF

* Radius of Curvature $\rightarrow \infty \Rightarrow \text{Centripetal Acc}^n = 0$.

* Curvilinear Motion: Motion of a particle or translation w/o rotation of a rigid body along curved path.

* More than 1-translational DOF

* Radius of Curvature is finite $\Rightarrow \text{Centripetal Acc}^n \neq 0$

3.1. Kinematics of Particles

3.1.1. Rectilinear Motion

- * Eq^{ns} of Motion
- * Relative Motion
- * Constrained / Dependent Motion

** Eq^{ns} of Motion:

1. Uniformly Accelerated Motion : {a = const.}

$$a = \frac{dv}{dt} \Rightarrow \int_U^V dv = a \int_0^t dt \Rightarrow \boxed{v = u + at}$$

$$v = u + at = \frac{ds}{dt} \Rightarrow \int_0^s ds = \int_0^t \{u + at\} dt \Rightarrow \boxed{s = ut + \frac{1}{2}at^2}$$

$$a = \frac{dv}{dt} = \frac{dv}{ds} \cdot \frac{ds}{dt} \Rightarrow \int_U^V v \cdot dv = a \int_0^s ds \Rightarrow \boxed{v^2 - u^2 = 2as}$$

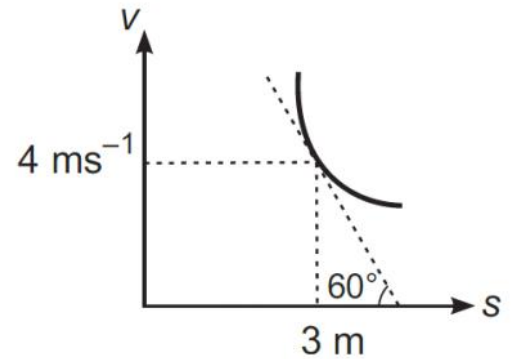
2. Non Uniformly Acc. Motion {a ≠ c}

a. $\boxed{v = \frac{ds}{dt}}$

b. $\boxed{a = \frac{dv}{dt}}$

c. $\boxed{a = v \cdot \frac{dv}{ds}} \Rightarrow \boxed{v \cdot dv = a \cdot ds}$

IM. A particle is moving along a straight line whose velocity-displacement graph is as shown in the figure. What is the magnitude of acceleration (in m/s^2) when displacement is 3 m ?

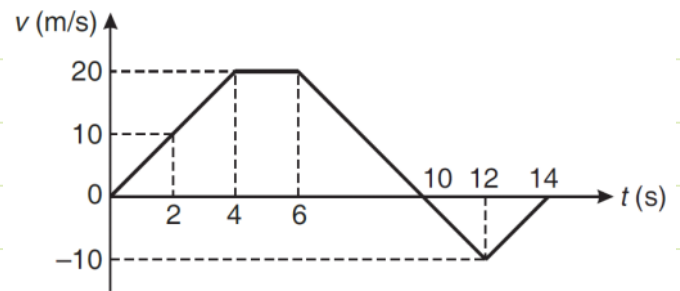


- (A) $2\sqrt{3}$
- (B) $3\sqrt{3}$
- ✓ (C) $4\sqrt{3}$
- (D) $5\sqrt{3}$

$$a = v \cdot \frac{dv}{ds} = 4 \{ \tan(120^\circ) \} = -4\sqrt{3} m/s^2$$

$$|a| = 4\sqrt{3} m/s^2$$

IM + IM. Velocity-time graph of a particle moving in a straight line is shown in figure. In the time interval from $t = 0$ to $t = 14$ s, find
(a) average velocity and
(b) average speed of the particle.



$$v = \frac{ds}{dt} \Rightarrow \int ds = \int v dt \Rightarrow \text{displacement} = \text{Area under } v.t \text{ curve.}$$

$$\text{displacement} = \left\{ \left(\frac{10+2}{2} \right) \times 20 \right\} - \left(\frac{1}{2} \times 4 \times 10 \right)$$

$$S = 100 \text{ m}$$

$$V_{av} = \frac{S}{t} = \frac{100}{14}$$

$$V_{av} = 7.14 m/s$$

$$\text{distance covered} = \left\{ \frac{1}{2} (10+2) \times 20 \right\} + \left\{ \frac{1}{2} \times 4 \times 10 \right\}$$

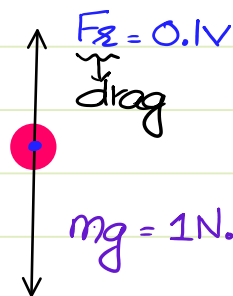
$$= 140 \text{ m.}$$

$$\text{speed}_{av} = \frac{140}{14}$$

$$\text{speed}_{av} = 10 m/s$$

2M xE

A particle of mass **0.1 kg**, which is released from rest, falls vertically downward under gravity in a fluid. The fluid offers a resistive force, which is linearly proportional to the particle velocity with **0.1 N.s/m** as the constant of proportionality. The uniform gravitational acceleration is **10 m/s²** throughout the trajectory of the particle. The magnitude of the particle velocity (in m/s) at time **1 s** after release (*rounded off to two decimal places*) is



$$\Sigma F = m \cdot a$$

$$\Rightarrow mg - 0.1v = m \cdot a$$

$$\Rightarrow a = g - v. \quad \{a = f(v)\}$$

$$\Rightarrow \frac{dv}{dt} = g - v.$$

$$\Rightarrow \int_0^1 dt = \int_0^v \frac{dv}{g-v}$$

$$\Rightarrow -1 = \ln(g-v) \Big|_0^v$$

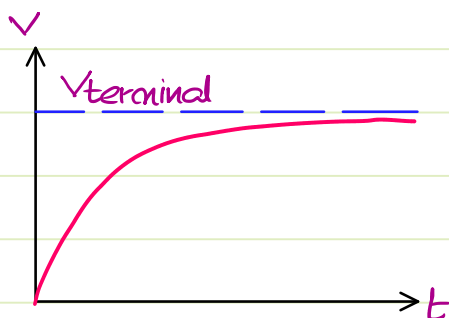
$$\Rightarrow -1 = \ln(g-v) - \ln(g)$$

$$\Rightarrow -1 = \ln\left\{\frac{g-v}{g}\right\}$$

$$\therefore \frac{g-v}{g} = e^{-1}$$

$$\therefore v = g\left\{1 - \frac{1}{e}\right\}$$

$$v = 6.321 \text{ m/s}$$



QM. $\therefore a-s$ graph is given & the problem is regarding v .

$$v \cdot dv = a \cdot ds = dA$$

$\therefore \int v \cdot dv = \text{Area under the curve.}$

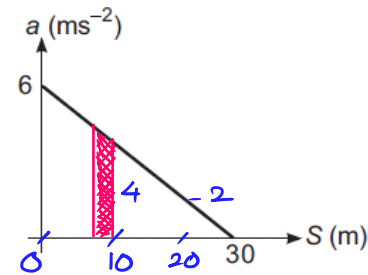
$$\Rightarrow \frac{v^2}{2} = \text{Area under the curve.}$$

from $s=0$ to $s=10$ m.

$$\text{Area} = \left\{ \frac{6+4}{2} \right\} \times 10 = 50$$

$$\therefore \frac{v^2}{2} = 50 \Rightarrow v = 10 \text{ m/s.}$$

A train starts from rest at $S = 0$ and is subjected to an acceleration as shown in figure.



- (a) Velocity at the end of 10 m displacement is 20 ms^{-1}
- (b) Velocity of the train at $S = 10 \text{ m}$ is 10 ms^{-1}
- (c) The maximum velocity attained by train is $\sqrt{180} \text{ ms}^{-1}$
- (d) The maximum velocity attained by the train is 15 ms^{-1}

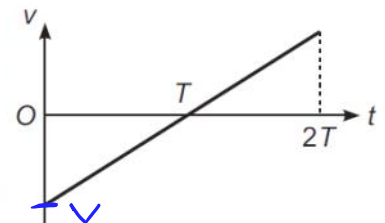
For Max velocity we have to find the Max area under curve

$$\frac{v_{\text{max}}^2}{2} = \frac{1}{2} (30 \times 6)$$

$$\Rightarrow v_{\text{max}} = \sqrt{180} \text{ m/s.}$$

QM. for direction of motion to change the sign of $\vec{v}$ must change $\therefore$ dirⁿ of motion changes at $t = T$.

The figure shows the velocity (v) of a particle plotted against time (t).



- (a) The particle changes its direction of motion at some point
- (b) The acceleration of the particle remains constant
- (c) The displacement of the particle is zero
- (d) The initial and final speeds of the particle are the same

$$a = \frac{dv}{dt} \quad \therefore v \propto t \quad \therefore \frac{dv}{dt} = \text{const} \quad \therefore a = \text{const}$$

$$\text{In } v-t \text{ graph } S = \text{Area under the curve} = \left\{ \frac{1}{2} TV - \frac{1}{2} TV \right\} = 0.$$

By similarity of Δ 's the initial & final velocities have same magnitudes.

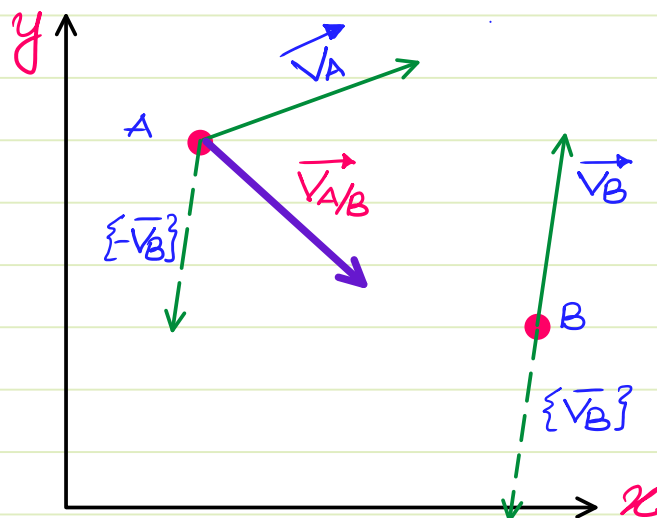
Relative Motion: The study of Motion of a body wrt to another moving body.

* The body wrt which the motion is being analysed, always assumes itself to be:

- origin
- at rest.

* $\therefore$ that body gives itself **Negative** of whatever motion vector it actually has.

* It then assigns the same negative vector to the other bodies & then observes them.



$$\vec{V}_{B/B} = \vec{V}_B + \{-\vec{V}_B\}$$

Abs. Motion
of "B"

The imagination
of B so as to
remain at rest.

Rel. velocity of B
wrt B = 0.

$$\vec{V}_{A/B} = \vec{V}_A + \{-\vec{V}_B\}$$

$$= \vec{V}_A - \vec{V}_B$$

$$\Rightarrow \vec{V}_A = \vec{V}_B + \vec{V}_{A/B}$$

for any motion parameter $\{\vec{k}: \text{vel, disp, acc}^n\}$

$$\vec{k}_{A/B} = \vec{k}_A + \{-\vec{k}_B\} = \vec{k}_A - \vec{k}_B$$

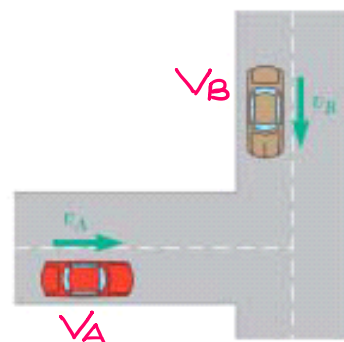
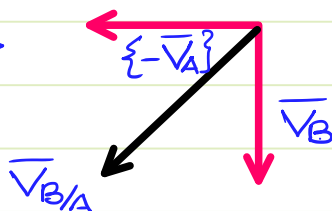
$$\text{||y } \vec{k}_A = \vec{k}_B + \vec{k}_{A/B}$$

1M.

Two cars are approaching an intersection at constant speeds as shown. What velocity will car B appear to have to an observer in car A?

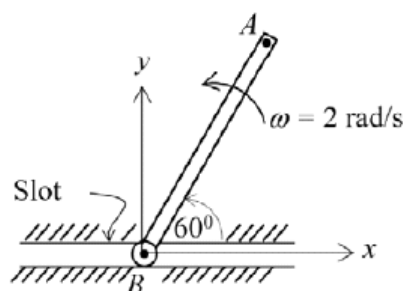
- (a) $\rightarrow$ (b) $\searrow$ (c) $\swarrow$ (d) $\nearrow$ (e) $\swarrow$

$$\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A$$



2M XE

End B of the 2 m long rigid rod AB is constrained to move horizontally in the slot as shown in the figure and has a velocity of $1.0 \hat{i}$ m/s. The angular velocity of the rod at the instant shown is 2 rad/s. The unit vectors along x and y directions are denoted respectively as $\hat{i}$ and $\hat{j}$. The velocity of point A in m/s is then given by

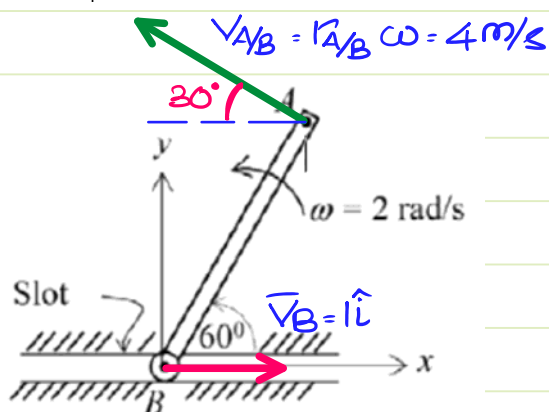


(A) $(1 - 2\sqrt{3})\hat{i} + 2\hat{j}$

(B) $(1 + 2\sqrt{3})\hat{i} - 2\hat{j}$

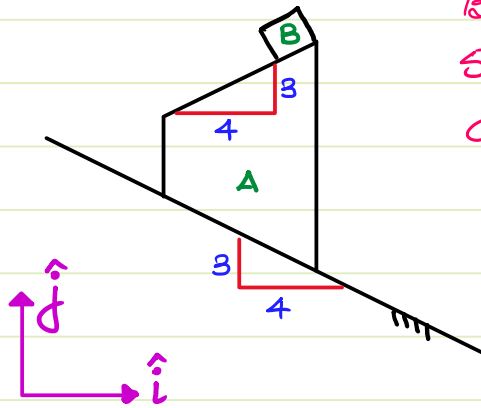
(C) $-2\sqrt{3}\hat{i} + 2\hat{j}$

(D) $2\sqrt{3}\hat{i} - 2\hat{j}$



$$\begin{aligned} \vec{v}_{A/B} &= -4 \cos 30^\circ \hat{i} + 4 \sin 30^\circ \hat{j} \\ &= -2\sqrt{3} \hat{i} + 2 \hat{j} \\ \vec{v}_A &= \vec{v}_{A/B} + \vec{v}_B \\ &= \{-2\sqrt{3} + 1\} \hat{i} + 2 \hat{j} \end{aligned}$$

IM.



Block A slides down the fixed incline at 5 m/s. It observes block B to be sliding down its incline at 10 m/s.

$$\vec{V}_B = \{ \underline{\hspace{1cm}} \hat{i} + \underline{\hspace{1cm}} \hat{j} \} \text{ m/s}$$

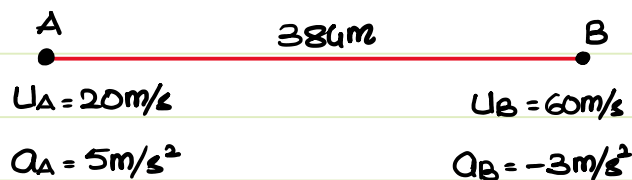
$$\vec{V}_A = \{ 4 \hat{i} - 3 \hat{j} \} \text{ m/s}$$

$$\vec{V}_{B/A} = \{ -8 \hat{i} - 6 \hat{j} \} \text{ m/s}$$

$$\vec{V}_B = \vec{V}_A + \{ \vec{V}_{B/A} \}$$

$$= \{ -4 \hat{i} - 9 \hat{j} \} \text{ m/s}$$

IM.



06. An auto A is moving at 20 m/s and accelerating at 5 m/s² to overtake an auto B which is 384 m ahead. If auto B is moving at 60 m/s and decelerating at 3 m/s², at what time would A pass B?

(a) 10 secs

(b) 16 secs

(c) 20 secs

(d) 25 secs

$$S_{A/B} = 384 \text{ m}$$

$$U_{A/B} = -40 \text{ m/s}$$

$$a_{A/B} = 8 \text{ m/s}^2$$

$$t = ?$$

$$S_{A/B} = U_{A/B} t + \frac{1}{2} a_{A/B} t^2$$

$$384 = -40t + \frac{1}{2} 8t^2$$

$$t = 16 \text{ s}, -6 \text{ s.}$$

-ve time represents past time

$$\therefore t = 16 \text{ s.}$$

Constrained Motion : Motion of a body depends on Motion of another body connected to it. $\therefore$ This motion is also known as Dependent Motion.

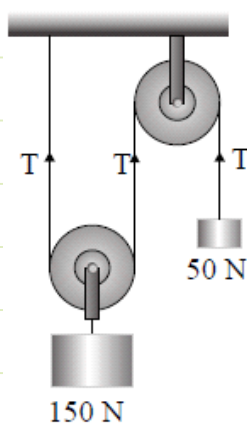
Steps In Analysing Constrained Motion :

1. The links connecting are assumed to be inextensible.
 $\Rightarrow$ The length of these links can be used to define the positions of the bodies wrt certain reference.

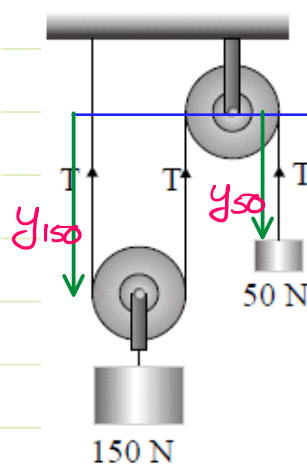
No. of eq^{ns} of constrain = No. of DOF of the system.

2. Apply mathematical operations on the constrain eq^{ns} to develop the eq^{ns} of Motion.

Mark.



Find the velocity of 150 N block if 50 N block moves up with a velocity of 6 m/s.



Ref. Level.

Eqⁿ of length of the rope :

$$2y_{150} + y_{50} + c' = l$$

$$\Rightarrow \boxed{2y_{150} + y_{50} = C} \text{ Constrain Eqⁿ}$$

By differentiation :-

$$2\dot{y}_{150} + \dot{y}_{50} = 0$$

$$\therefore \dot{y}_{150} = -\dot{y}_{50}/2$$

$\dot{y}_{50} = -6 \text{ m/s}$ { $\because \dot{y}_{50}$ is decreasing the value of y_{50} }

$$\therefore \dot{y}_{150} = -\frac{-6}{2} = 3 \text{ m/s.}$$

$$\therefore \text{150 N block} = 3 \text{ m/s}$$

Eqⁿ of constrain : $x_A + 3y_B = C$

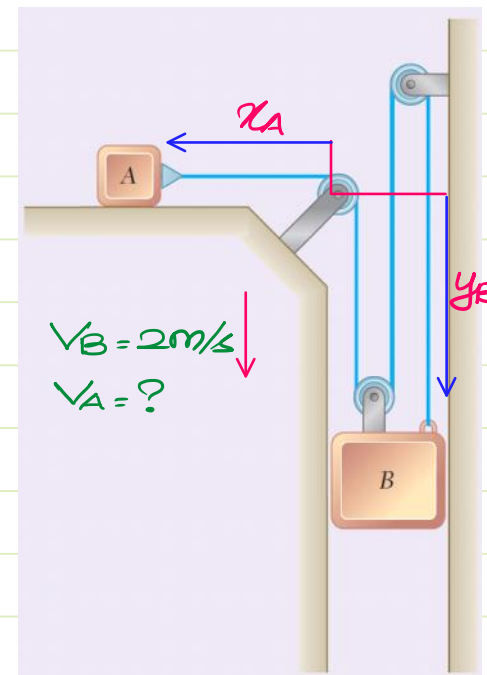
By diff: $\dot{x}_A + 3\dot{y}_B = 0$

$$\dot{y}_B = 2 \text{ m/s}$$

$$\Rightarrow \dot{x}_A + 3\{2\} = 0$$

$$\Rightarrow \dot{x}_A = -6 \text{ m/s}$$

$$\Rightarrow V_A = 6 \text{ m/s} \rightarrow$$

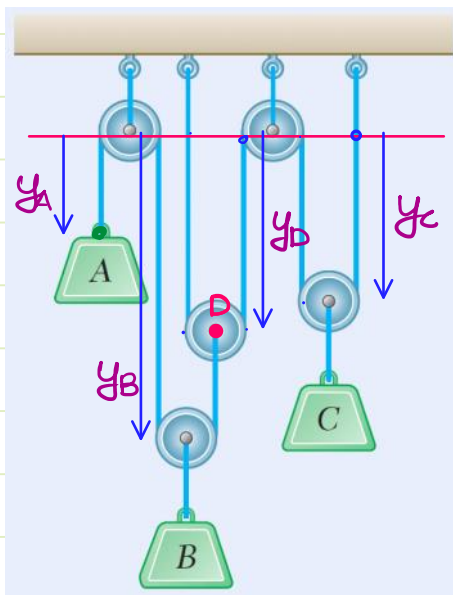


2M.

What should be the velocity of block A so that the block B is at rest while C moves down at 3m/s?

$\therefore$ the system has 2 ropes $\therefore$ No. of DOF = 2

$\therefore$ No. of eqⁿs of const. = 2.



Eq₁:

$$2y_C + 2y_D = C$$

$$\Rightarrow y_C + y_D = C_1$$

$$\text{Eq}_2: y_A + y_B + y_B - y_D = C_2$$

$$y_A + 2y_B + y_C - C = C_2$$

$$\Rightarrow y_A + 2y_B + y_C = C.$$

$$\Rightarrow \dot{y}_A + 2\dot{y}_B + \dot{y}_C = 0$$

$$\dot{y}_A + 3 = 0 \Rightarrow \dot{y}_A = -3 \text{ m/s}$$

$\therefore$ Vel of block A = 3 m/s $\uparrow$

